

CSE 331

points to submit

shoily hoy ma'am

L> attendance - 10

Quiz = 20

Mid = 25

Final = 35

Assignment = 100 points : 5 groups

Automata

↓
automatic / self controlled

Symbol : a, b, c, 0, 1, 2, \$, *

Σ = alphabet. $[|\Sigma| = 2 \rightarrow \text{size of alphabet}]$

word for automata is "string" বাক্য)

Language : string এর length on string
কিছু কিছু বাক্য থাকবে তা হল
Language

* Number of string = $| \Sigma |^n \rightarrow$ length of language
 $\downarrow$
 size of alphabet.

$$\Sigma = \{a, b, c, d\}$$

$$| \Sigma | = 4$$

language: string of length 5 (n)

$$\text{number of strings} = | \Sigma |^n$$

$$= 4^5 = 1024$$

Power of alphabet:

$$\text{alphabet } \Sigma = \{0, 1\}$$

$\Sigma^0 = \emptyset / \{ \}$... empty set or power zero

string of zero length $\Sigma^0 = \emptyset / \{ \}$

cardinality = no. of element/string in language.

string of 2 length $\Sigma^2 = \{0,1\} \{0,1\}$

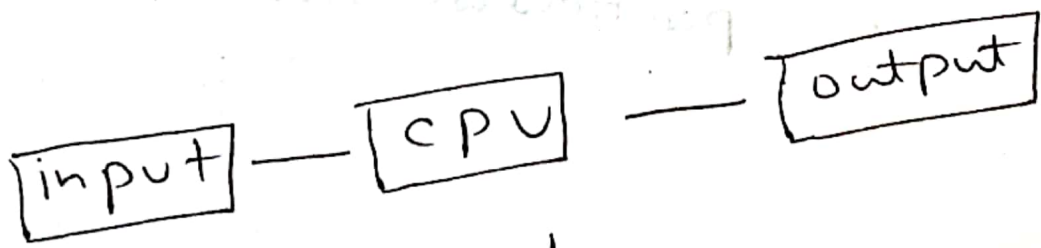
$$\Sigma \cdot \Sigma = \{00, 01, 10, 11\}$$

Σ^* = string of 0 on any length.

$$\Sigma^* = \emptyset, 0, 1, 01, 00$$

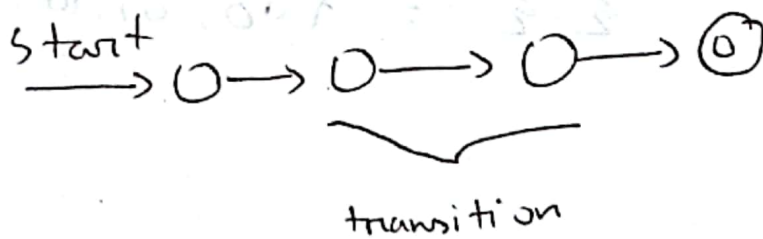
Σ^+ (string of any length except zero)

$$\begin{aligned}\Sigma^+ &= \Sigma \cdot \Sigma^* \\ &= (0,1) \cdot (\emptyset) \\ &= (0,1)\end{aligned}$$



theoretical model
start $\rightarrow 0$

transition & state



final state

①

(बस्यारि 210 वा 211)

DFA = Deterministic Finite automata

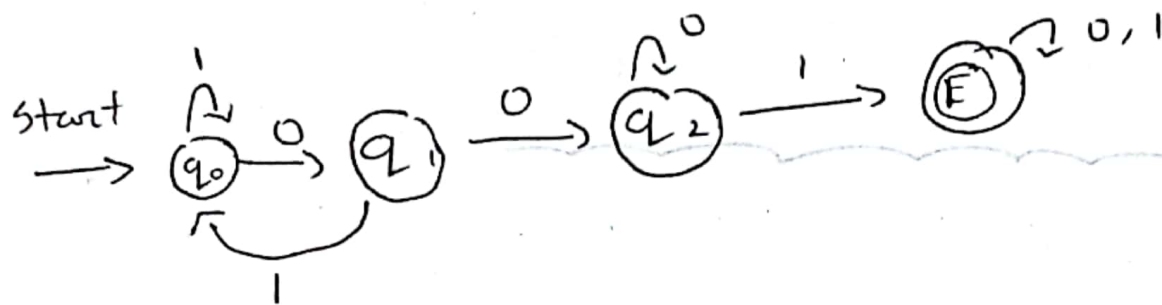
NFA = Non-deterministic Finite automata

Finite automata

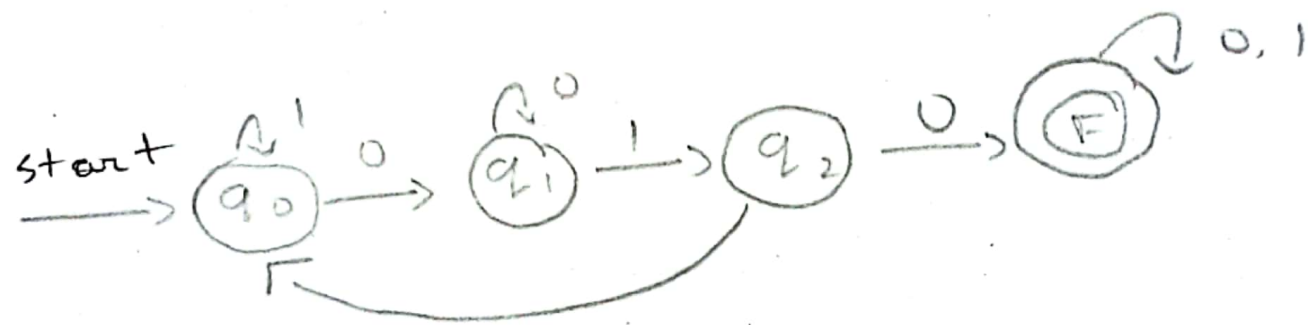
works with in particular state.

DFA

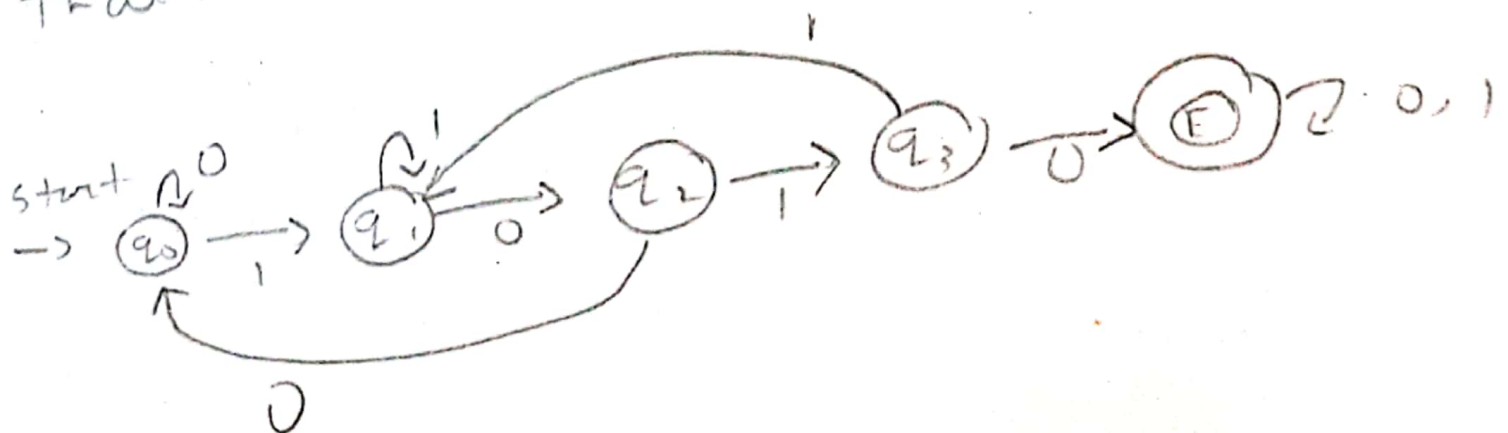
string that contains 001



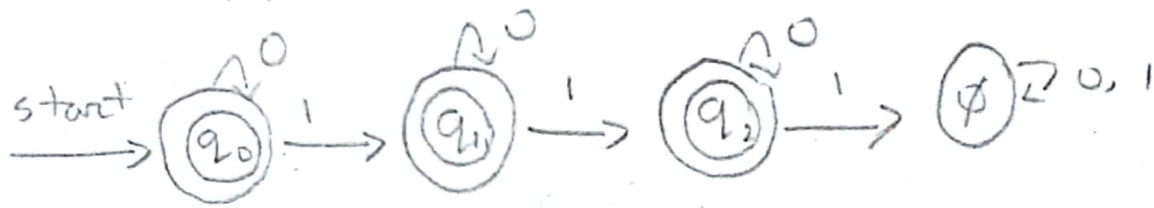
string that contains 010



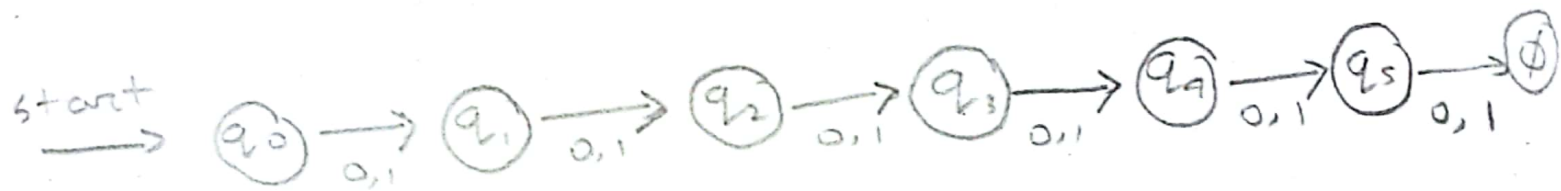
that contains 1010



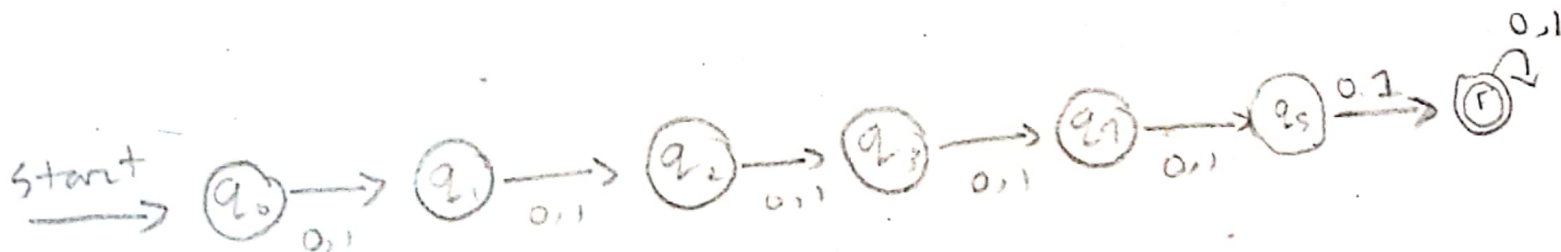
at most 2 (1) s



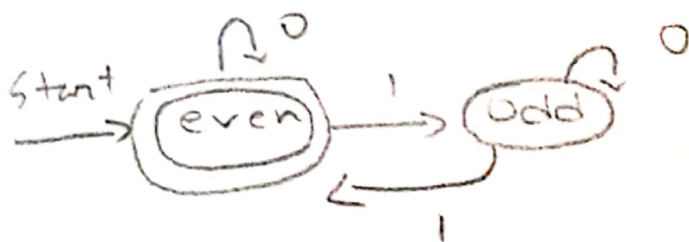
string having length exactly 5 alphabet



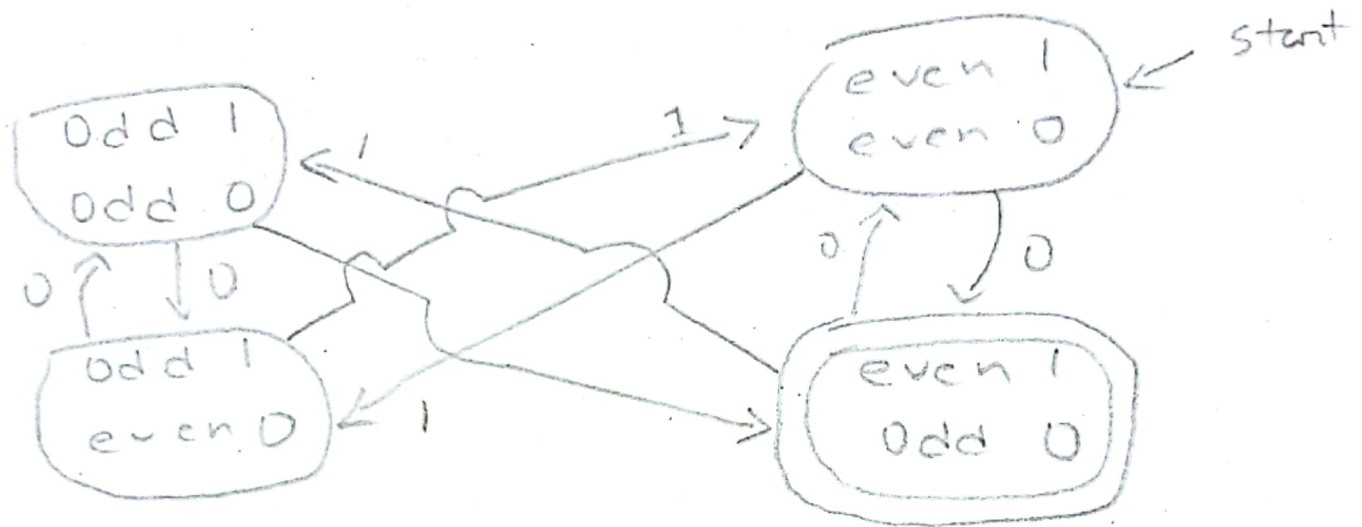
at least 5 alphabet



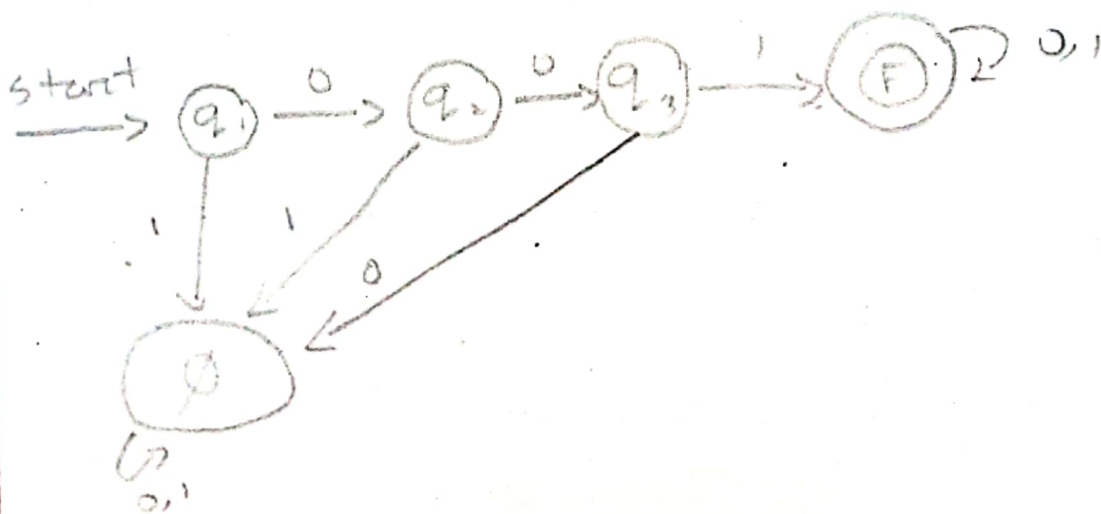
Only even number of 1



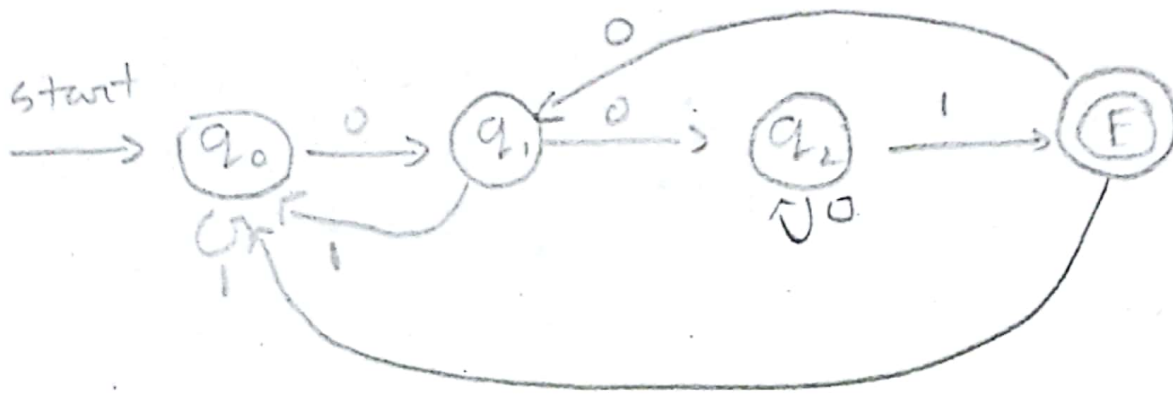
even number of 1 and odd number of
Zero



string that starts with 001



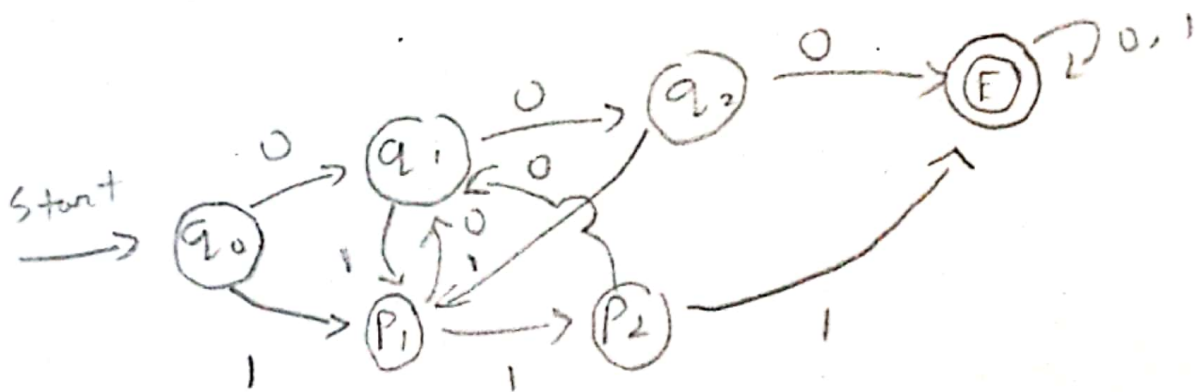
ends with 001



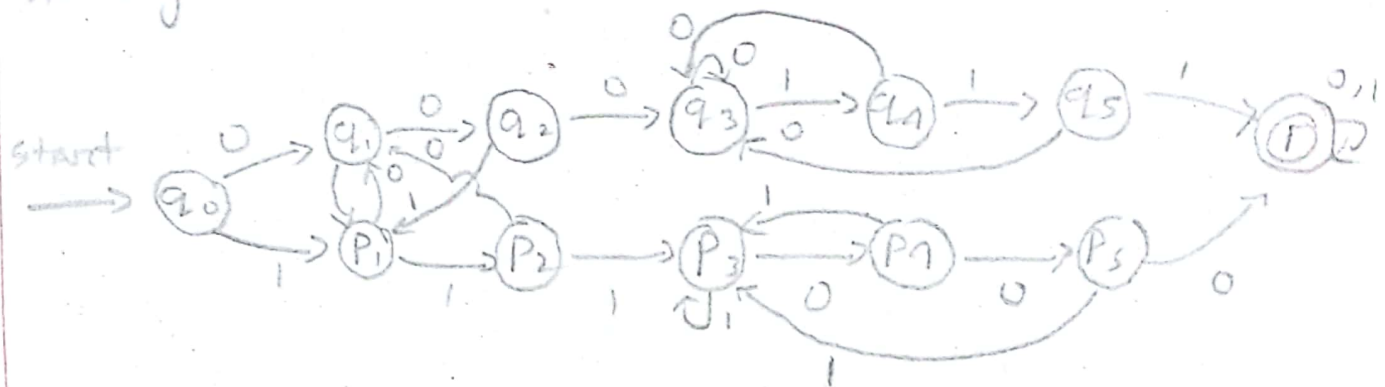
(20) 5 tracks binary strings

["on" $\Rightarrow$ Branch, "And" $\Rightarrow$ all possible output]

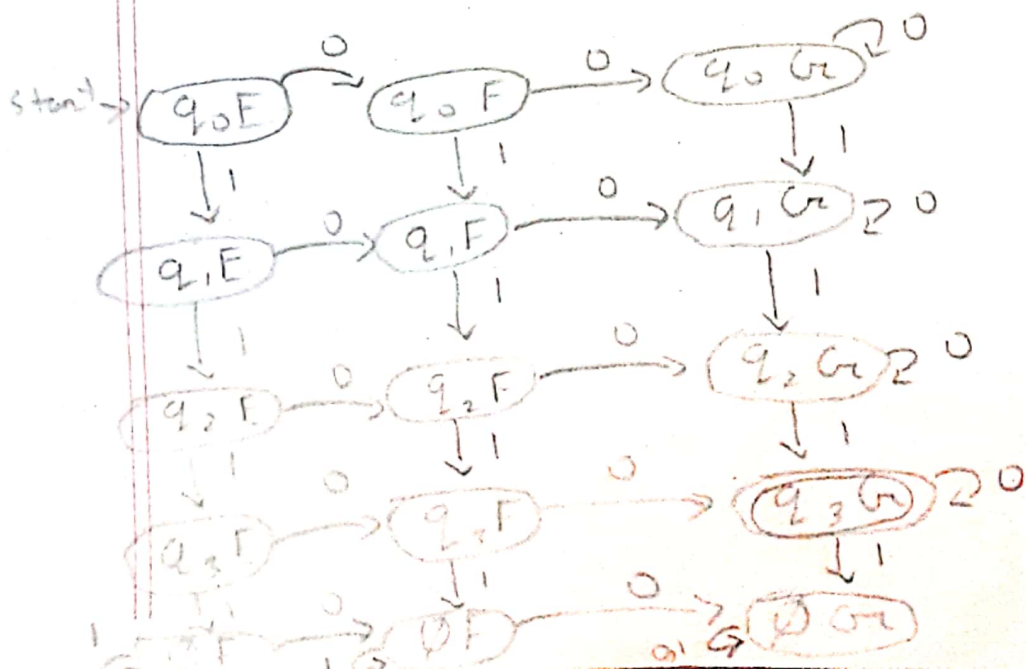
string that contains 000 or 111



string that contains 000 and 111



exactly 3 1's and atleast 2 (0's)

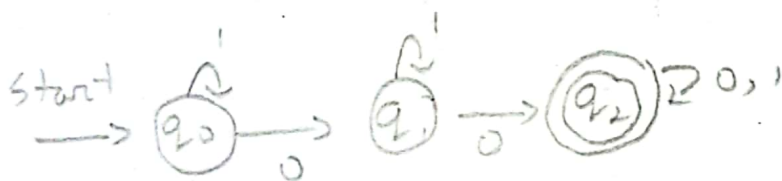


* DFA

- transition diagram
- Transition table
- Formal defⁿ

$$\Sigma = \{0, 1\}$$

at least 2 0's



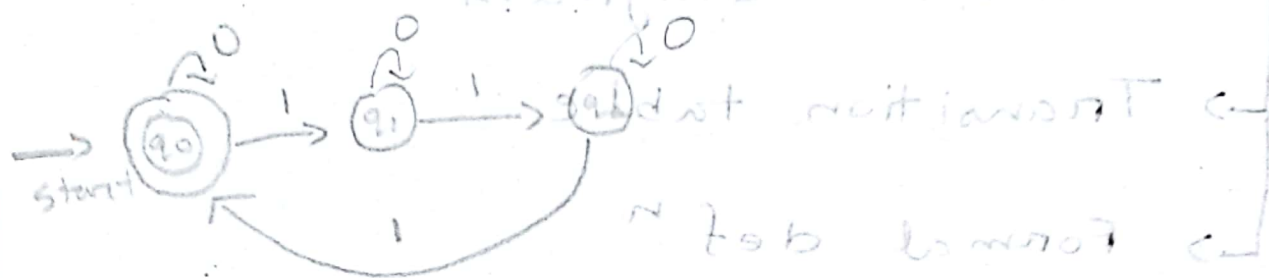
transition table

state	0	1
q ₀	q ₁	q ₀
q ₁	q ₂	q ₁
q ₂ *	q ₂ *	q ₂ *

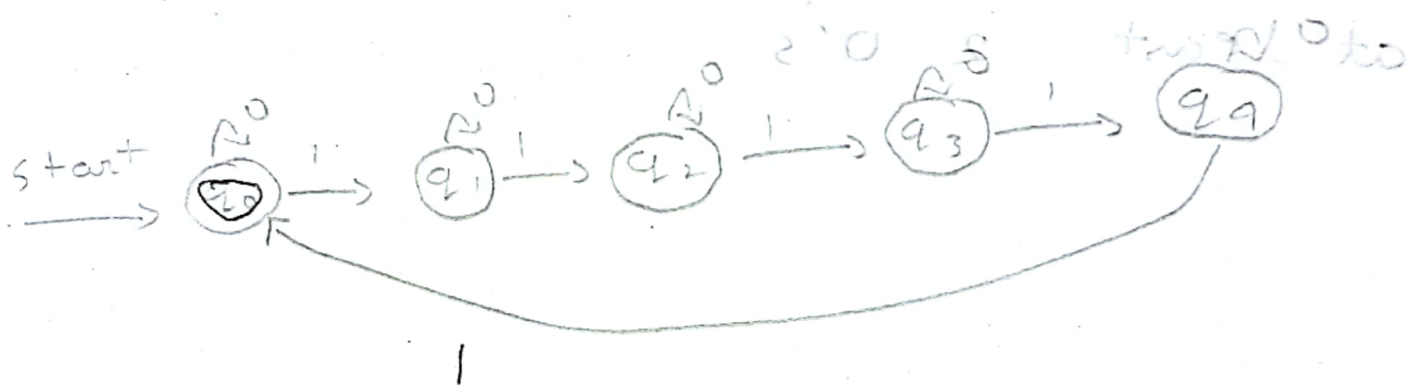
[* Final state q₂]

[new state q₂ added to the table]

1's divisible by 3



1's divisible by 5



Formal Defⁿ of DFA

1. Q (set of states)
2. Alphabet / Σ
3. Transition function / δ
4. start state / q_0
5. final state / $*$ / f

DFA (S) = (Q, Σ , δ , q_0 , final)

5 tuples

1. $Q = \{q_0, q_1, q_2, q_3, q_4\}$

2. $\Sigma = \{0, 1\}$

4. start state = q_0

5. final state = q_4

transition function

current state $\rightarrow$ input $\rightarrow$ next state
 $\delta(q_0, 0) = q_0$

$\delta(q_0, 1) = q_1$

$\delta(q_1, 0) = q_1$

$\delta(q_1, 1) = q_2$

$\delta(q_2, 0) = q_2$

$\delta(q_2, 1) = q_3$

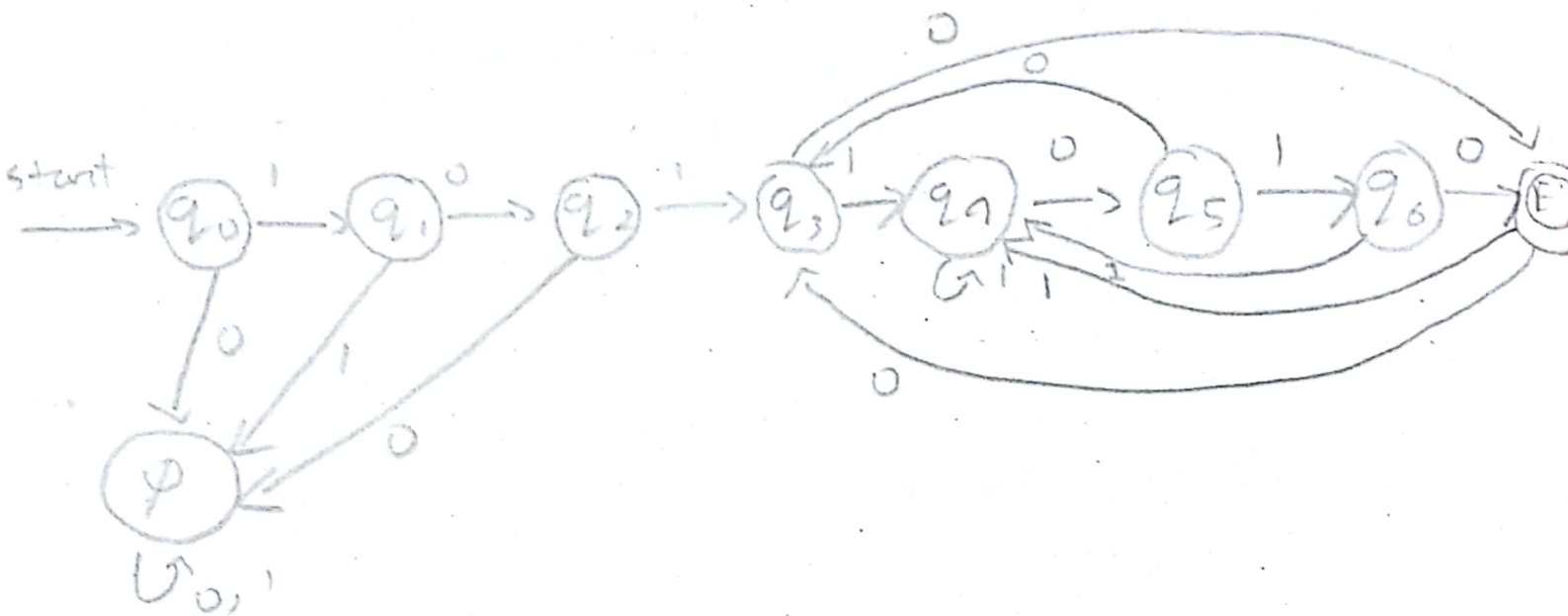
$\delta(q_3, 0) = q_3$

$\delta(q_3, 1) = q_4^*$

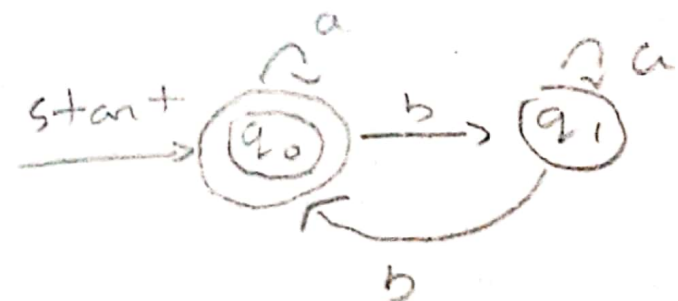
$\delta(q_4, 0) = q_4^*$

$\delta(q_4, 1) = q_4^*$

starts 101 and end 1010



DFA that starts with a on accepts even number of b's.

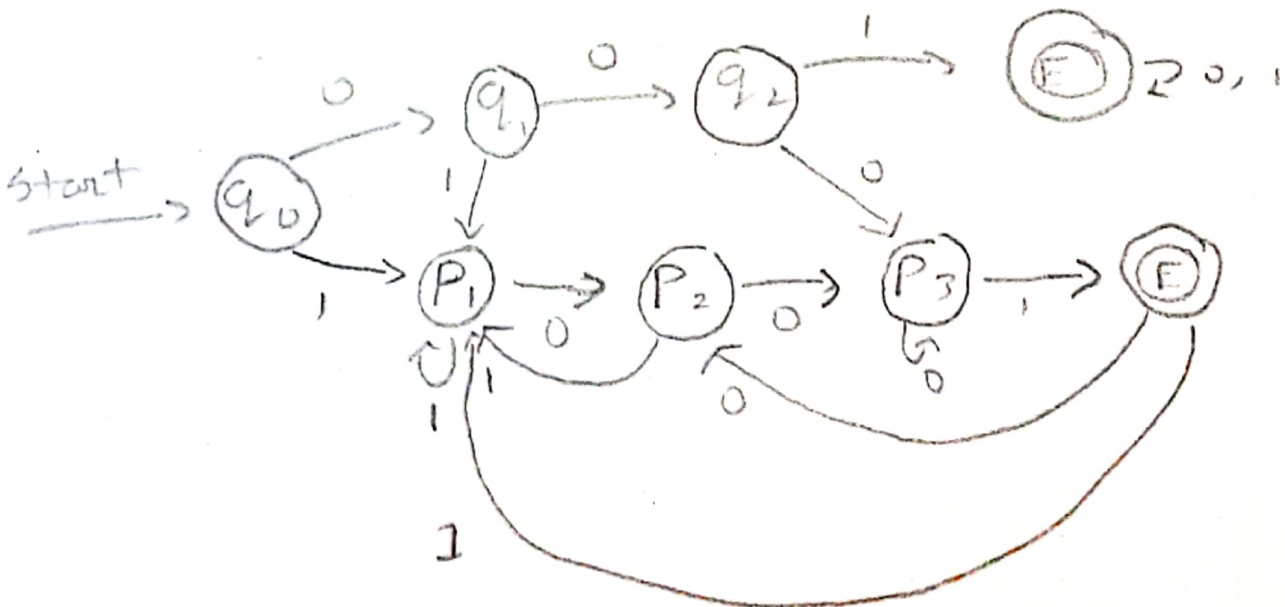


[on morn = brunch 227]

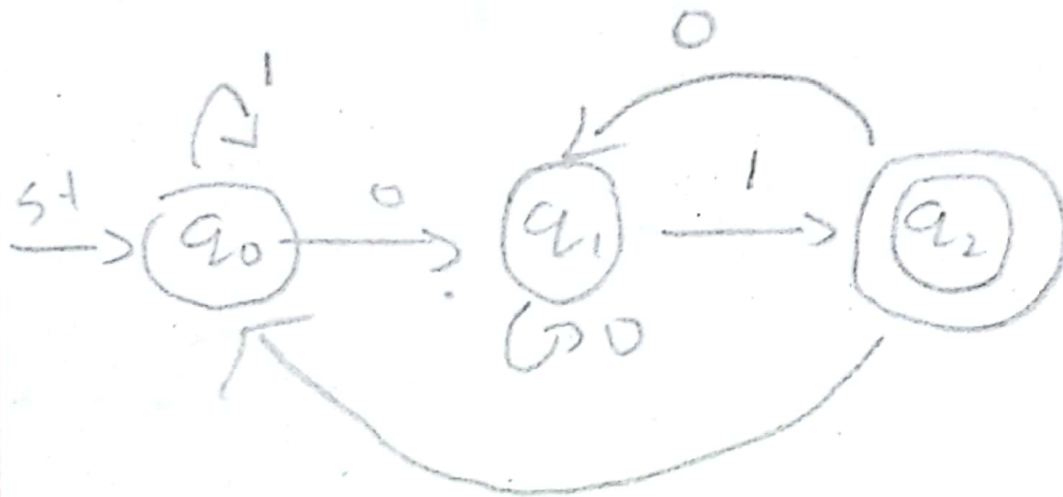
[and $2\pi\alpha =$ All possible way]

[and ନିମ୍ନ ଲିଖିତ କାର୍ଯ୍ୟ ସମ୍ପାଦନ କର]
 ନିମ୍ନ ଲିଖିତ କାର୍ଯ୍ୟ ସମ୍ପାଦନ କର
 ନିମ୍ନ ଲିଖିତ କାର୍ଯ୍ୟ ସମ୍ପାଦନ କର]

starts with 001 or ends with 001



ends with 01

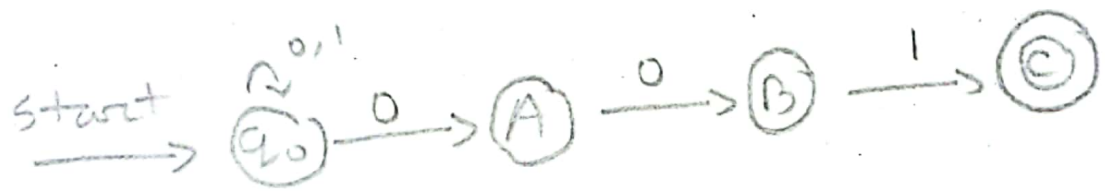


NFA

NFA

→ ଏହା input ଓ output (ଉପର ଉଲ୍ଲେଖ) ନାହିଁ।

end with 001

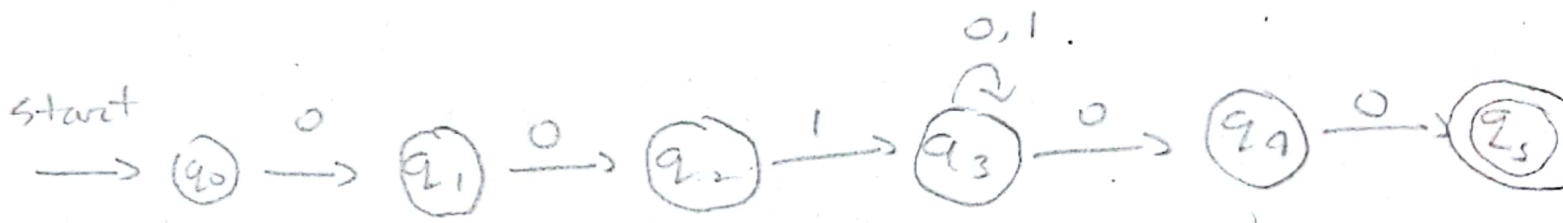


starts with 001

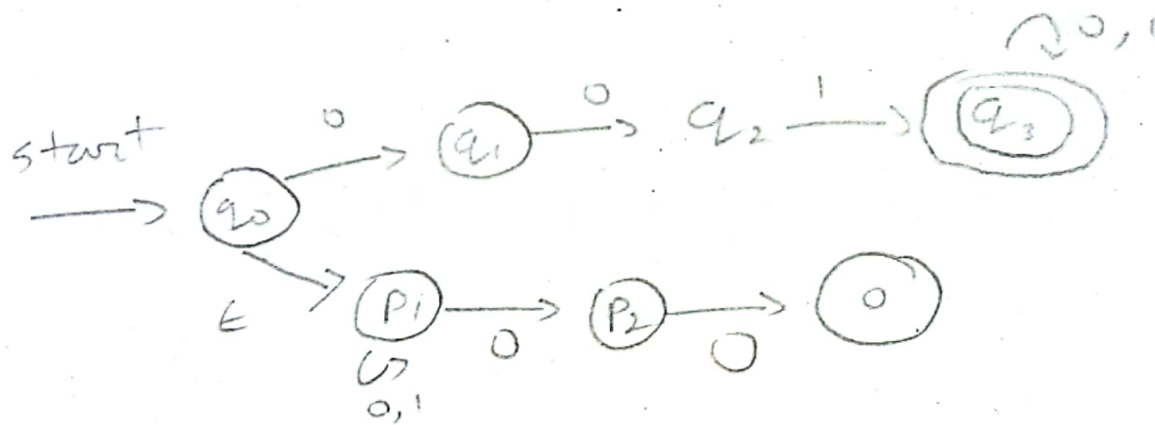


[ଯେତେବେଳେ NFA ର ଉପରକୁ DFA
କରି ଦିଅ]

start 001 and end 00



start with 001 or end with 00

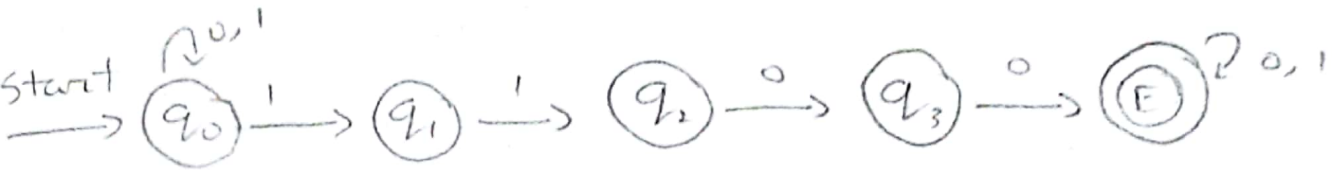


[NFA ko start, end, contain state]

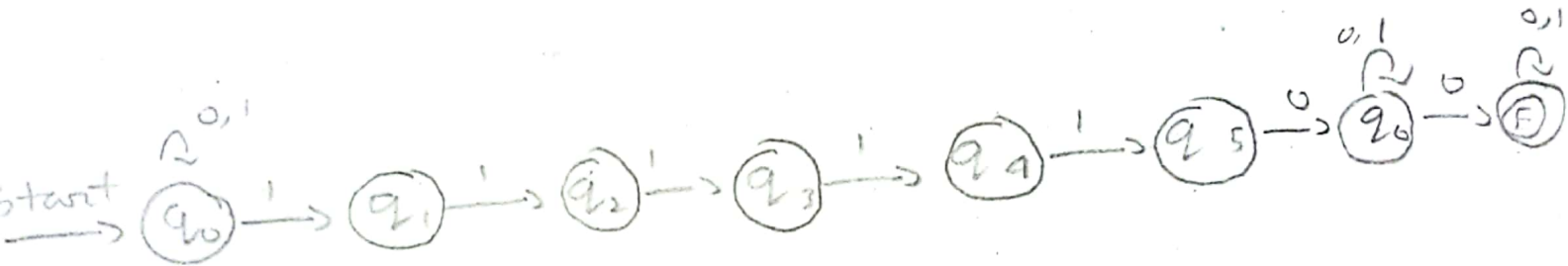
jo state hai wo NFA ko wo state hai

[NFA ko DFA]

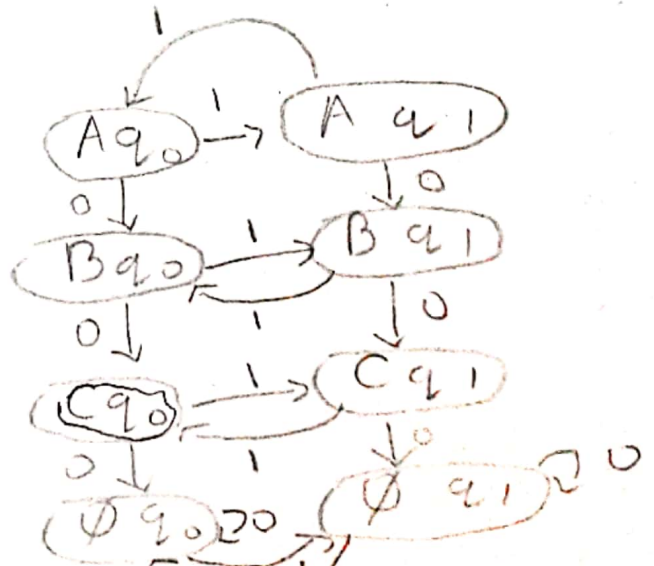
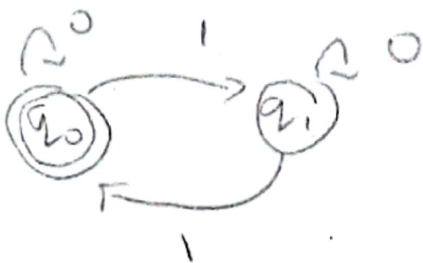
Contain 1100



5 consecutive 1's followed by atleast 2 0's



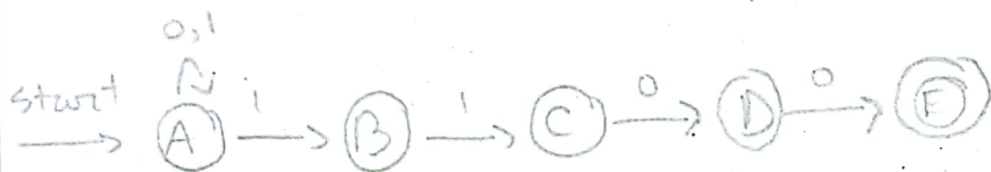
even number of 1's and exactly 2 0's.
 → DFA



Formal definition of NFA

NFA $(Q, \Sigma, \delta, \text{start state}, \text{final state})$

NFA ends with 1100



$Q = \{A, B, C, D, E\}$

$\Sigma = \{1, 0\}$

start = A

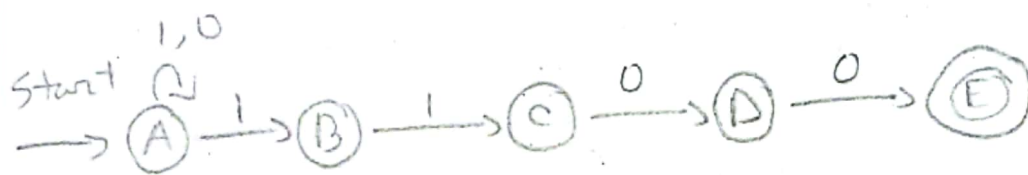
Final = E

Transition function δ

$\delta(A, 0) = A$	$(C, 1) = \emptyset$
$\delta(A, 1) = B, A$	$(D, 0) = E^*$
$\delta(B, 0) = \emptyset$	$(D, 1) = \emptyset$
$\delta(B, 1) = C$	$(E^*, 0) = \emptyset$
$\delta(C, 0) = D$	$(E^*, 1) = \emptyset$

↓ [NFA ਤੋਂ DFA conversion
using subset construction]

[NFA ਤੋਂ DFA ਤੋਂ ਡਾਇਗਰਾਮ ਬਣਾਓ ਅਤੇ
transition table ਬਣਾਓ]



state	0	1
A	A	A, B
A, B	A	A, B, C
A, B, C	A, D	A, B, C
A, D	A, E*	A, B
A, E*	A	A, B

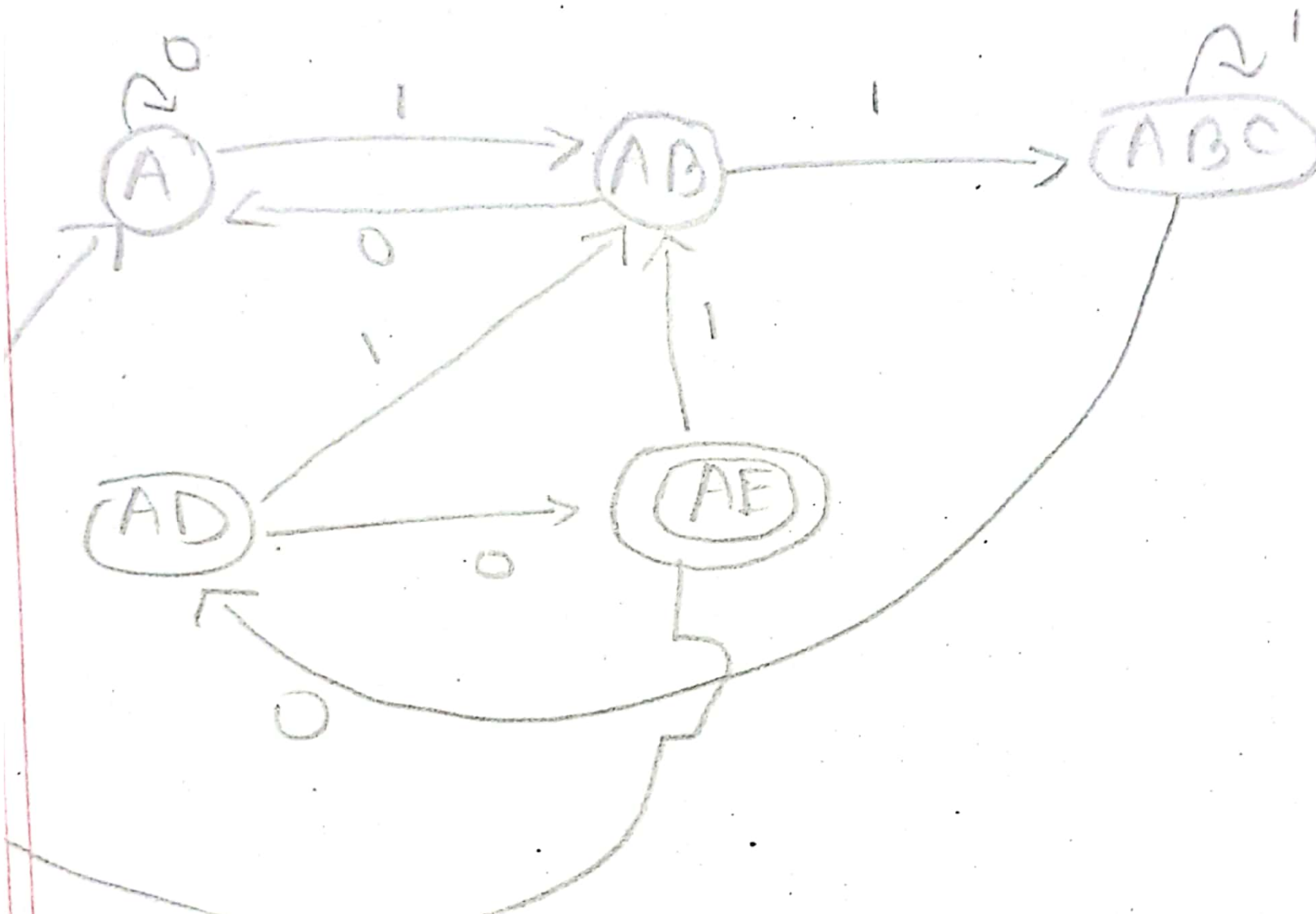
[state, $\emptyset$ ਆਮਤ
ignore $\emptyset$]

[ਘੁੰਮਣਾ $\emptyset$ ਆਮਤ
ਨਹੀਂ]

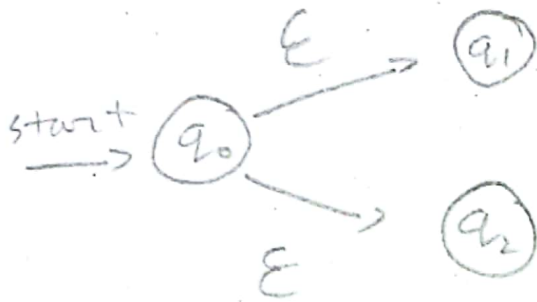
[ਜੇਕਰ letter
ਪ੍ਰਵੇਸ਼ਤ ਹੋਵੇ ਤਾਂ]

[ਜੇਕਰ ਹੋਰ ਨਹੀਂ
New ਡਾਇਗਰਾਮ]

DFA 76 ଜନାମ 3.3 0



Convert DFA or NFA to a single "ε" state
 using the concept of "Epsilon NFA" or



↓

ε closure of $q_0 = \{q_0, q_1, q_2\}$

ε closure of $q_1 = \{q_1\}$

[No state is reached from Afsilon
 closure]

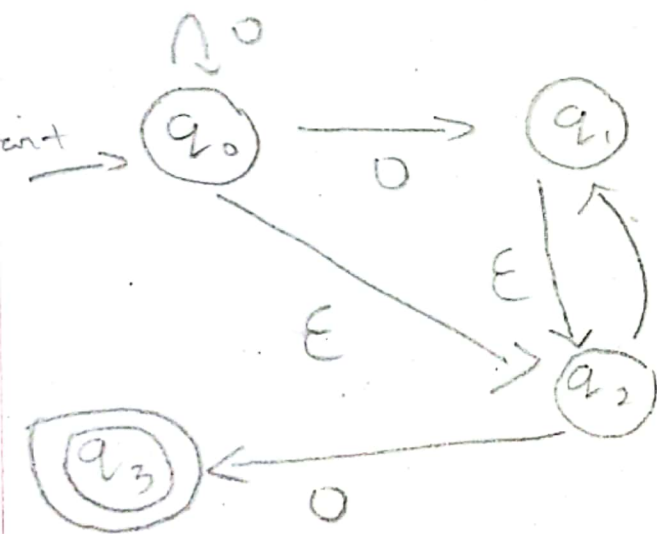
ϵ NFA (afsidon NFA) ରୁ DFA

ରୂପାନ୍ତରଣ

1. Afsidon closure find out

2. transition table

3. DFA ଗଠନ



ϵ - closure

$$q_0 = q_0, q_2$$

$$q_1 = q_1, q_2$$

$$q_2 = q_2$$

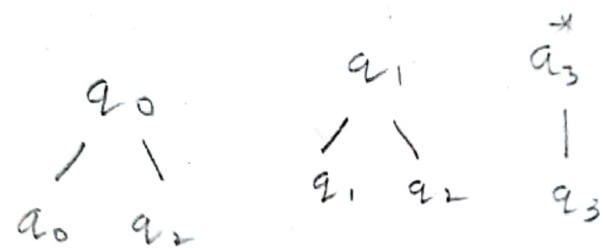
$$q_3 = q_3$$

[ଯଦି state ଥିବ
Afsidon ଥିବ ସମସ୍ତ
state ଠାରୁ ଯାଏ ତେବେ
ସେହି Afsidon closure]

transition table

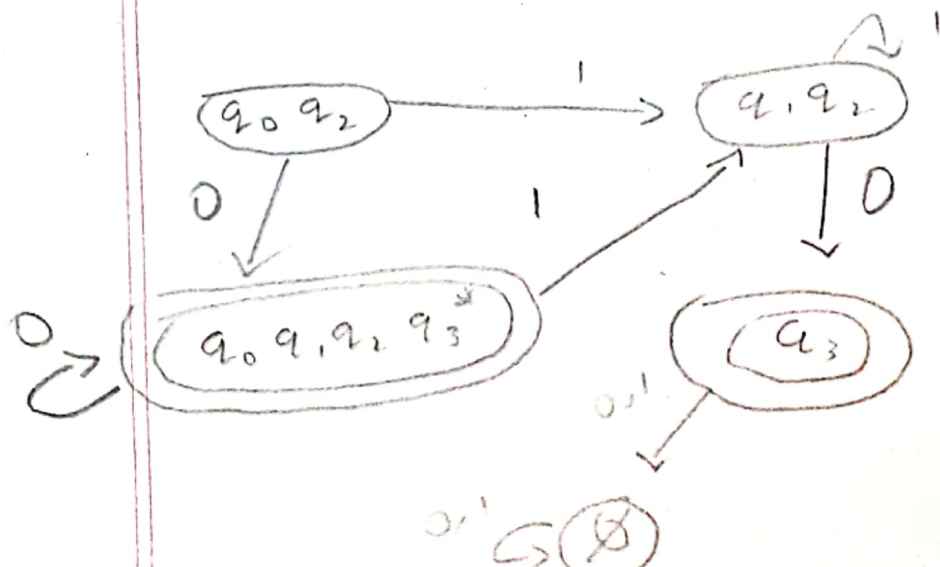
state	0	1
$q_0 \rightarrow q_0, q_2$	q_0, q_1, q_3^*, q_2	q_1, q_2
q_0, q_1, q_2, q_3^*	q_1, q_0, q_3^*, q_2	q_1, q_2
q_1, q_2	q_3^*	q_1, q_2
q_3^*	$\emptyset$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$

[start state is
Addition closure
for start
state]

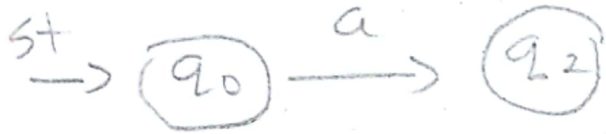


q_0, q_1, q_2, q_3^*

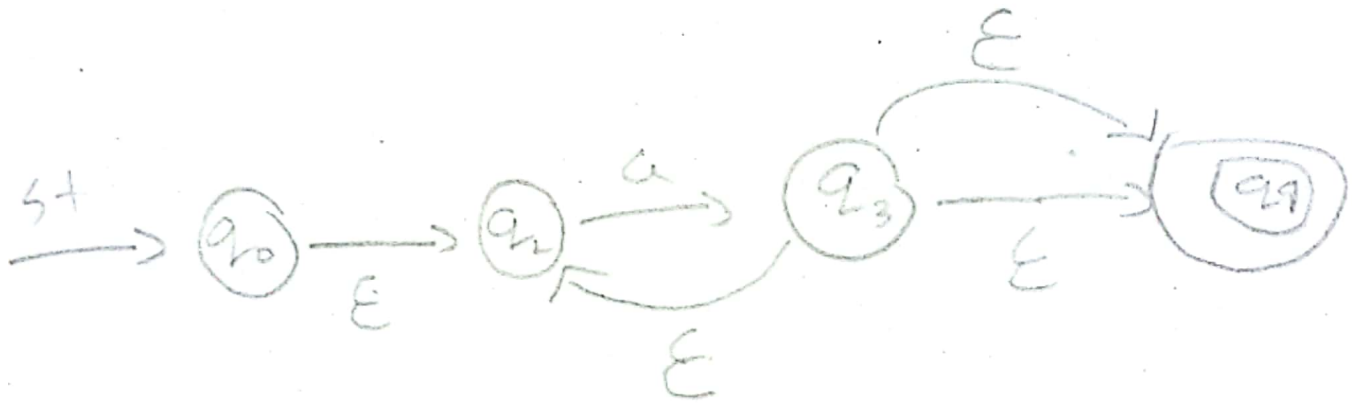
DFA



[Normal NFA to epsilon NFA]



↓



CSE 331

RE = Regular Expression / Regular Language

→ FA. ~~FA~~

Represent ~~not~~
LH ~~0 1 0 1 0 1~~
RL ~~0 1 0 1~~

[~~0 1 0 1~~] RE Operation
↓
()

* → Kleene closure → $a^* \Rightarrow \epsilon, a, aa, \dots$

+ → positive closure → $a^+ \Rightarrow a, aa, aaa, \dots$

· → concatenation → $a \cdot b \Rightarrow ab$

+ → $a + b \Rightarrow a, b$

{ } → $\{ a, b \}$
() → $(a + b)$

$$0.1^* = 0 \{ \epsilon, 1, 11, 111, \dots \}$$

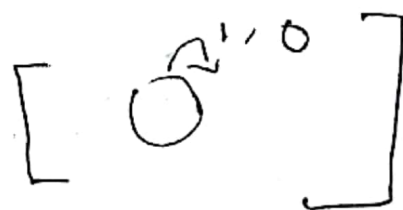
$$= 01, 011, 0, 0111$$

$$0.1^+ = 01, 011, \dots$$

$$1^*01^* + 10$$

$$\rightarrow 10, 101, 11011, 1011, \dots$$

$$*(1+0)^*$$



all combo of 1, 0

$$= \epsilon, 0, 1, 111, 000, \dots$$

* (star) નોડ પર self loop type બને છે.

$$(0.1)^*$$

$$= \{ \epsilon, 01, 0101, 010101, \dots \}$$

(0 and 1 duitai counted)

$\Rightarrow (1+0)^*$ binary strings of which starts

$\downarrow$
ends with one (1).

$10^*(0+1)01$

$= \boxed{1, 11, 01, 0001, \dots}$ strings of which starts
 $10^* + 10^*01$

$\Rightarrow$

$1(1+0)^*$

$\downarrow$ starts with 1

$(1+0)^* = \Sigma^*$

$[(1+0)^* \Sigma^*]$ alphabet Σ contains 0 and 1

$[(1+0)^* 1] \rightarrow$ shortest

contains 101

$= (1+0)^* 101 (1+0)^*$
 $= \Sigma^* 101 \Sigma^*$

starts with 10 and ends with 01

$10(1+0)^*01$

starts with 10 ^{OR} ~~ends~~ ends with 01

$10\varepsilon^* + \varepsilon^*01$

number of a exactly 1?
 $\downarrow$
 how many starts?

$\rightarrow b^* a b^*$

at least a number of 1?

$\rightarrow (a+b)^* a^+ (a+b)^*$

at most 2 a?

$\rightarrow b^* (a+\varepsilon) b^* (a+\varepsilon) b^*$

at least 2 0's on exactly 3 1's

$$(0+1)^* 0 (0+1)^* 1 0 (0+1)^* + 0^* 1 0^* 1 0^* 1 0^*$$

string that has even length

$$(\epsilon \epsilon)^* + (0 1 0)^* (0 1 0 1 0 1 0)^* \leftarrow$$
$$((0+1) (0+1))^*$$

string that divisible by 3

$$(\epsilon \epsilon \epsilon)^*$$

String that has odd length

$$\rightarrow \epsilon (\epsilon \epsilon)^*$$

string that not divisible by 3

$$\rightarrow (\epsilon \epsilon \epsilon)^* (\epsilon + \epsilon \epsilon)$$

number of 1's not divisible by 3.

$$\rightarrow (0^* 1 0^* 1 0^* 1 0^*)^* 0^* (1+0)^*$$

number of 1's not divisible by 3

$$(0^* 1 0^* 1 0^* 1 0^*) (0^* 1 0^* + 0^* 1 0^* 1 0^*)$$

set of strings consisting of even
number of a's followed by odd
number of b's

$$(a a)^* (b b)^* b$$

does not contain 01
 $1^* 0^*$

string in which every 0 is immediately followed by 11

$$(011+)^*$$

should have atleast one 0 and atleast one 1

$$\rightarrow (0+)^* 0 (0+)^* 1 (0+)^* + (0+)^* 1 (0+)^* 0 (0+)^*$$

starts and ends with a

$$a + a(a+b)^* a$$

start and ends with same symbol

$$a + a(a+b)^* a + b + b(a+b)^* b$$

start and ends with diff symbol

$a(a+b)^*b + b(a+b)^*a$

length atmost 3

$(a+b+c)^3$

length atleast 3

$(a+b)^3(a+b)^*$

string of odd length

$(\epsilon\epsilon)^*\epsilon$

string of even length

$(\epsilon\epsilon)^*$

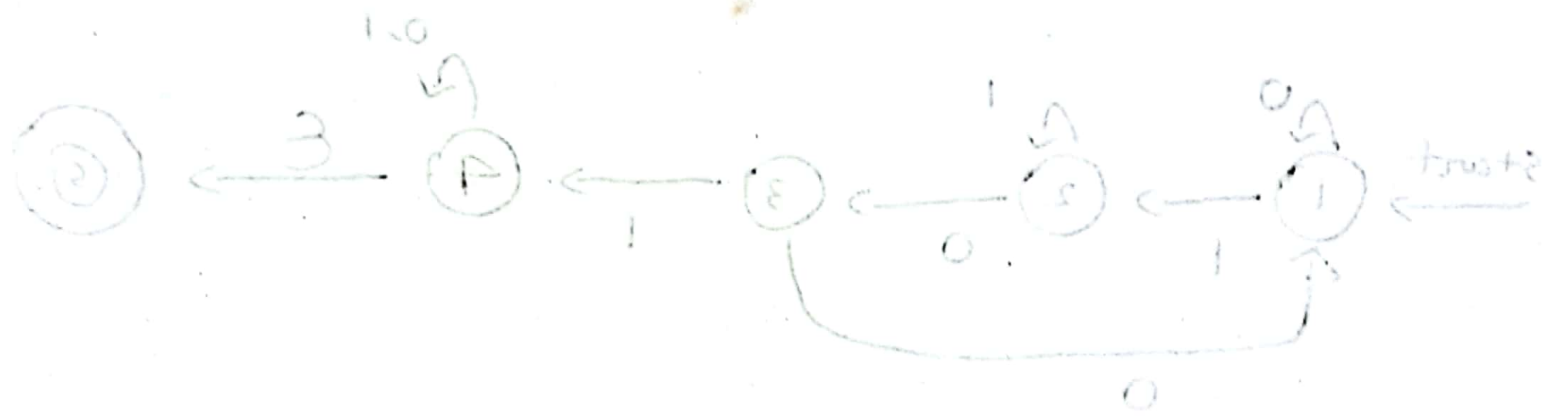
even number of 0

$(1^*01^*01^*)^*$

Odd number of 0's or 1's

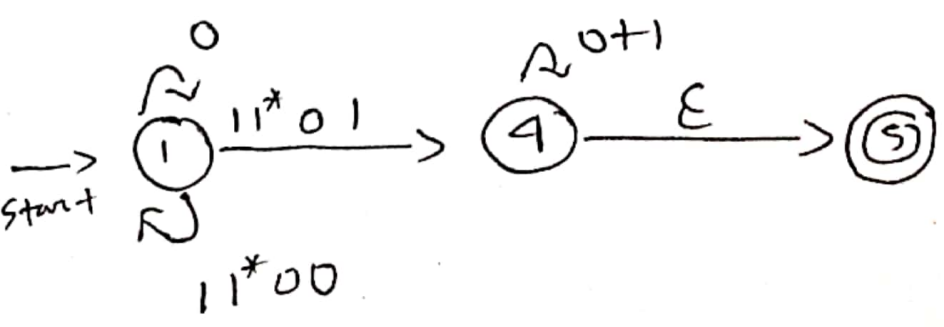
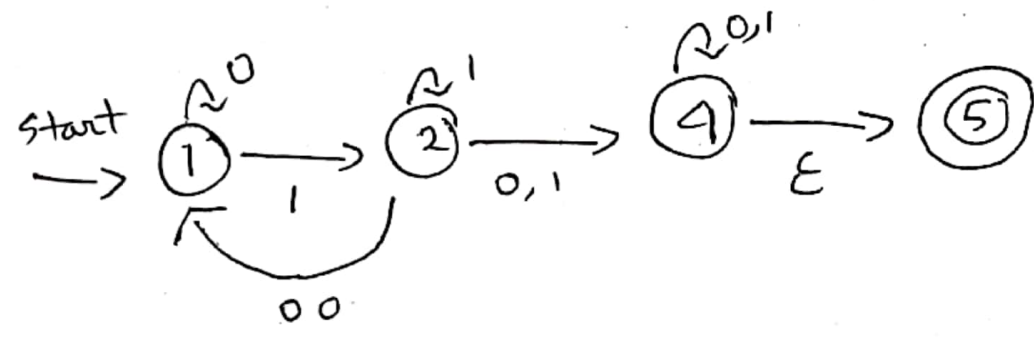
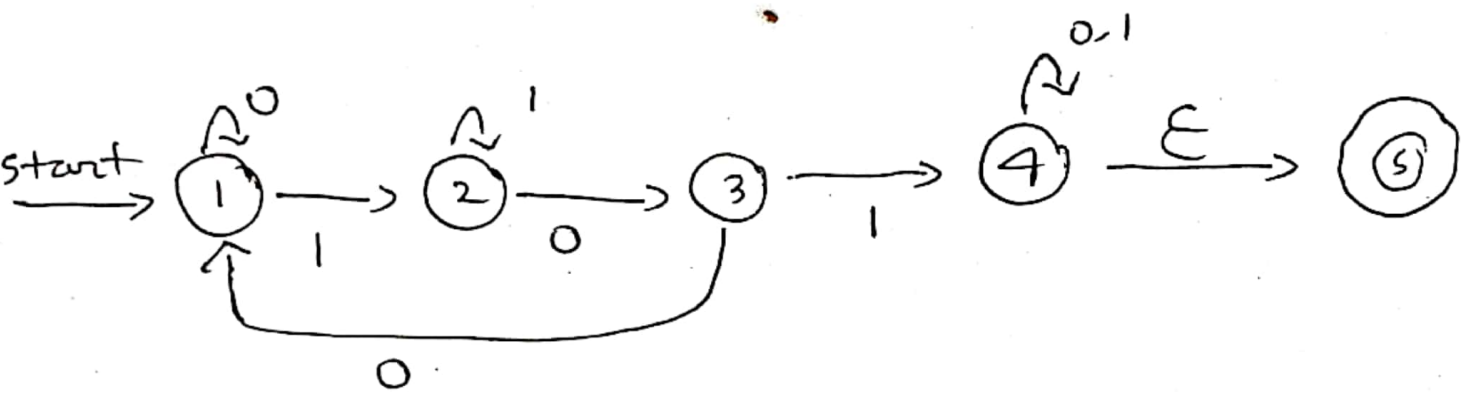
$(1^* 0 1^* 0 1^*)^* 0 1^*$

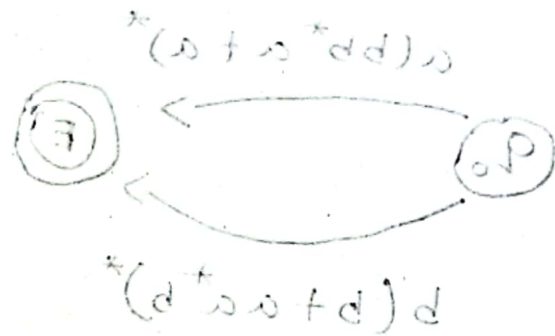
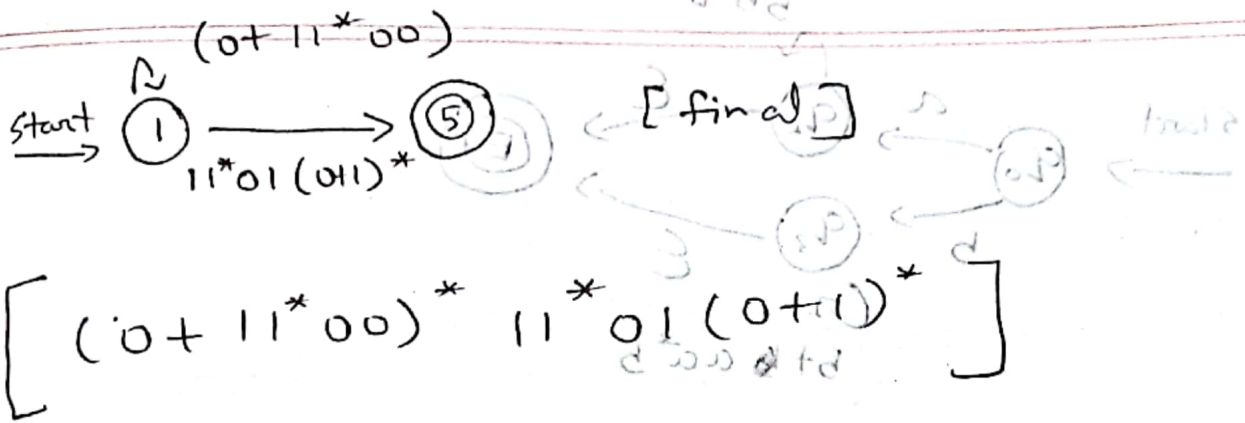
1. Start with state 0
 2. Remove multiple states
 3. If multiple final states, mark them as final states



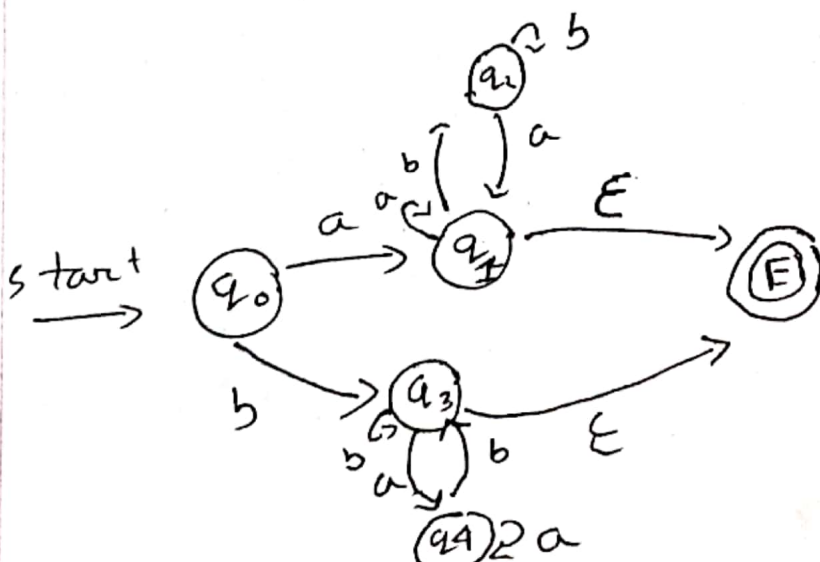
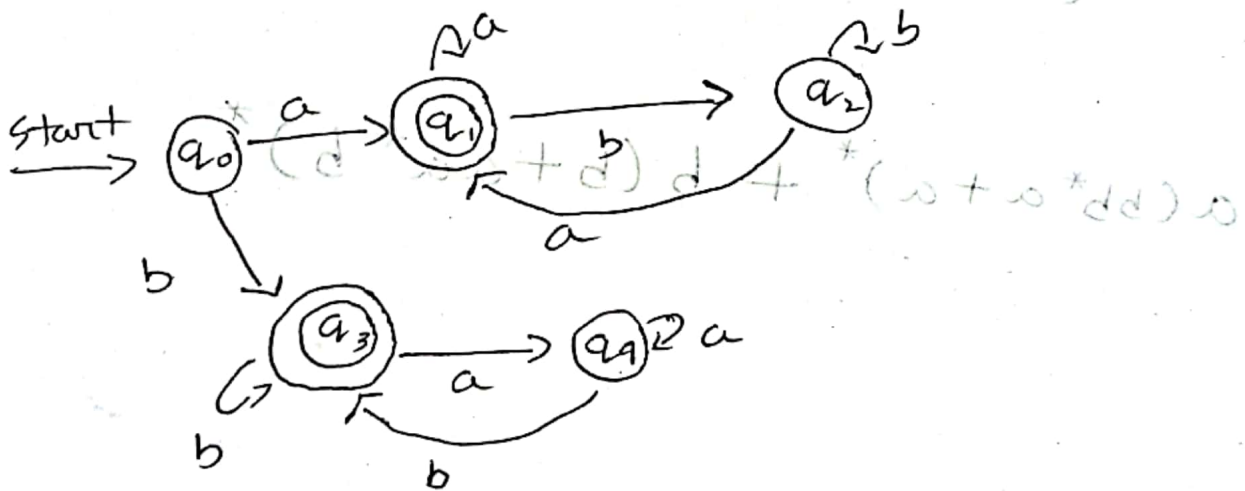
NFA / DFA to RE

1. start, final state $\rightarrow$ no change
2. Remove middle states.
3. if multiple final found, make 1 final state.

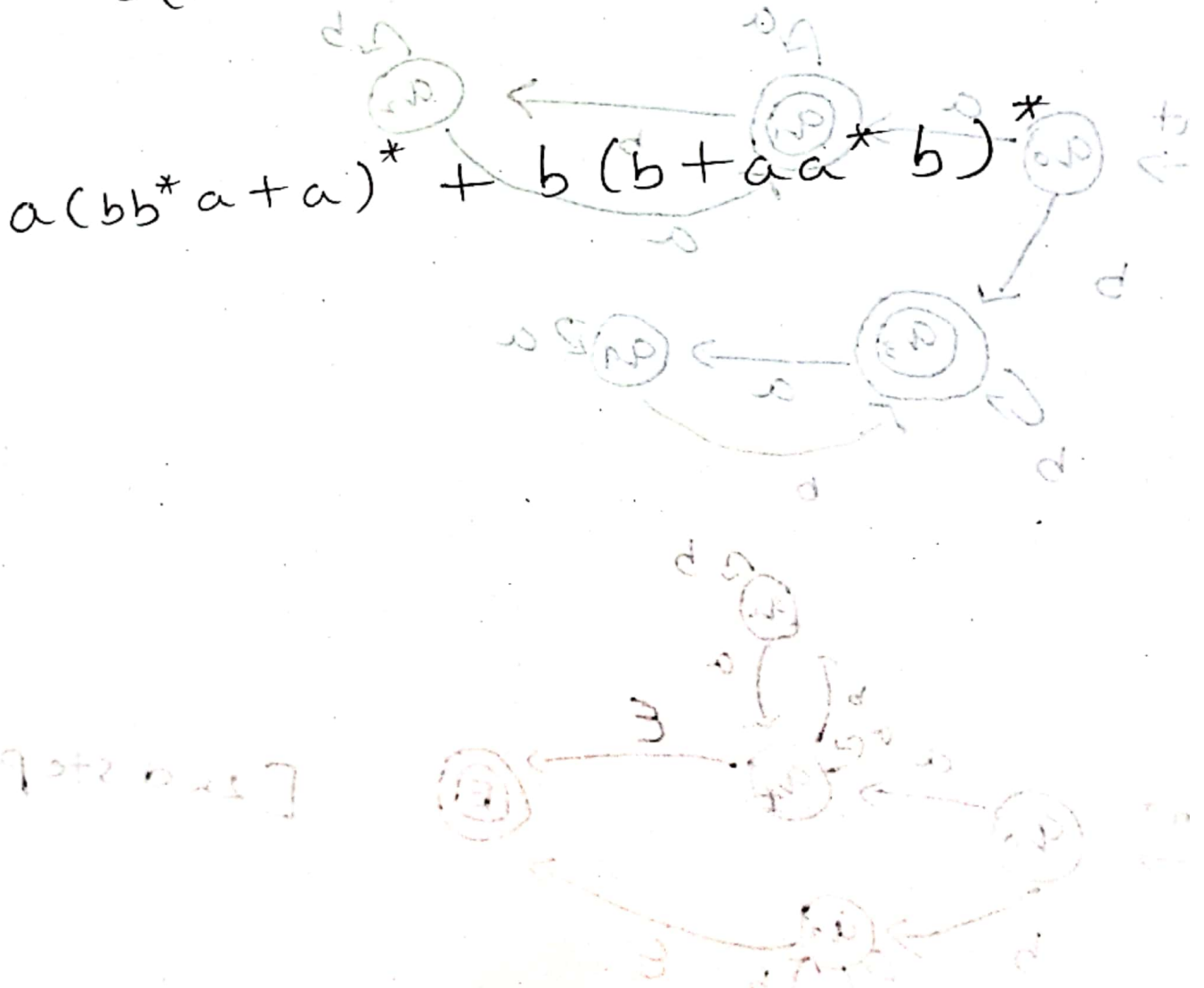
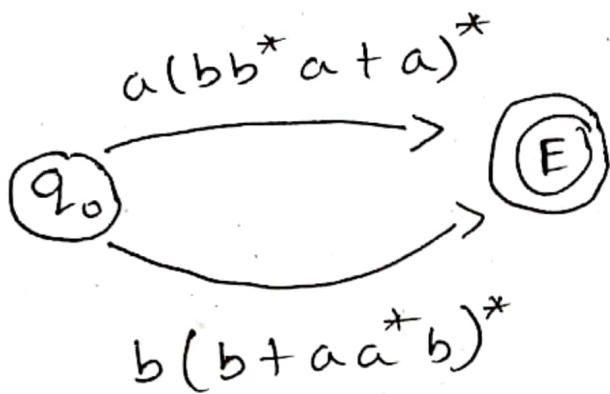
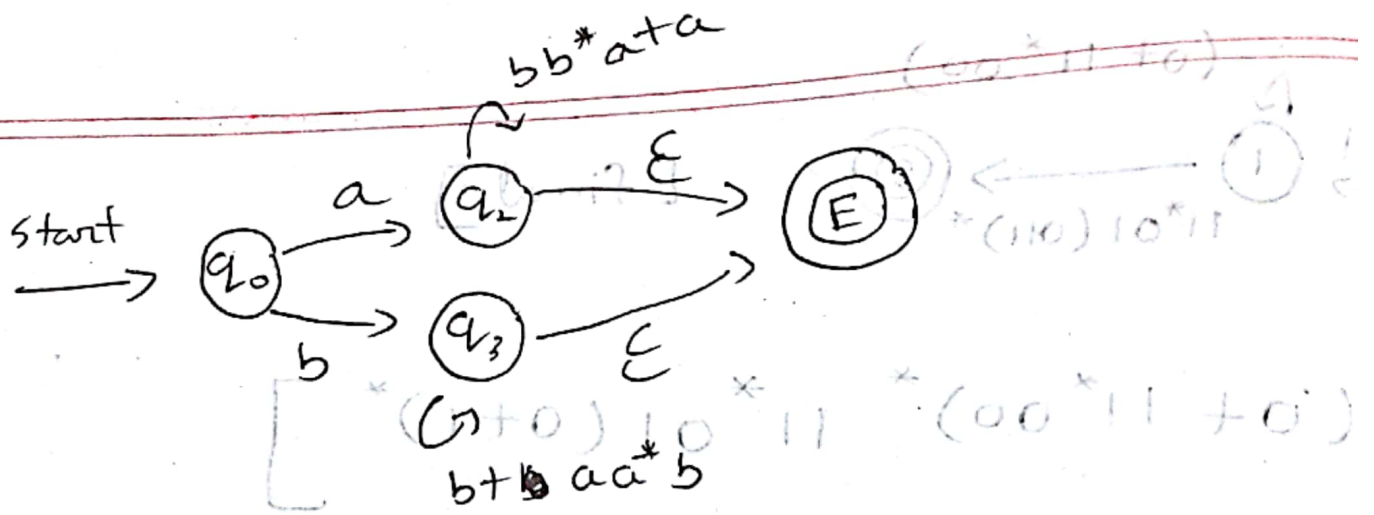




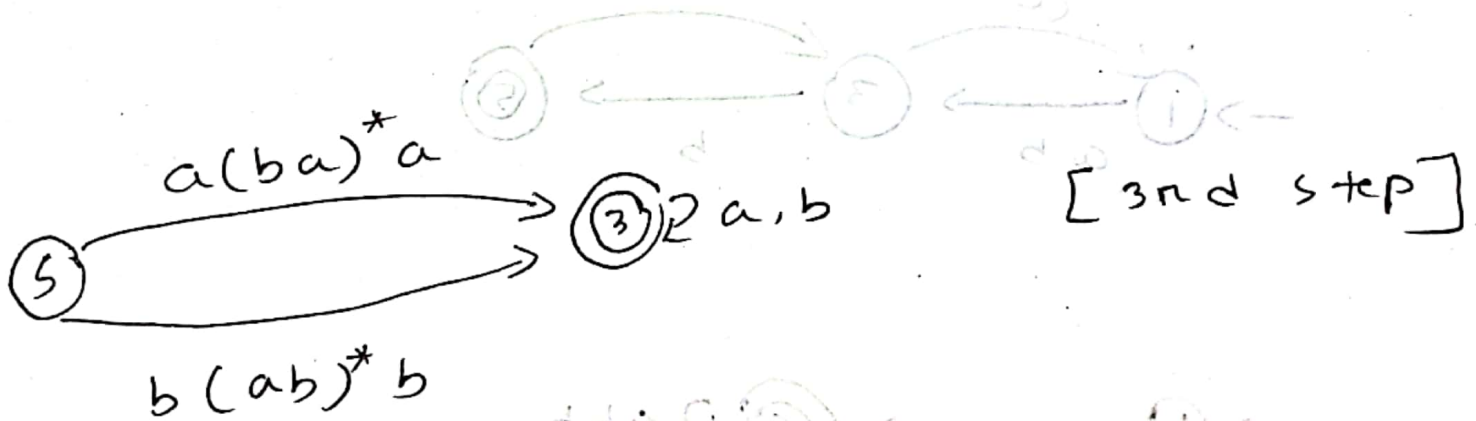
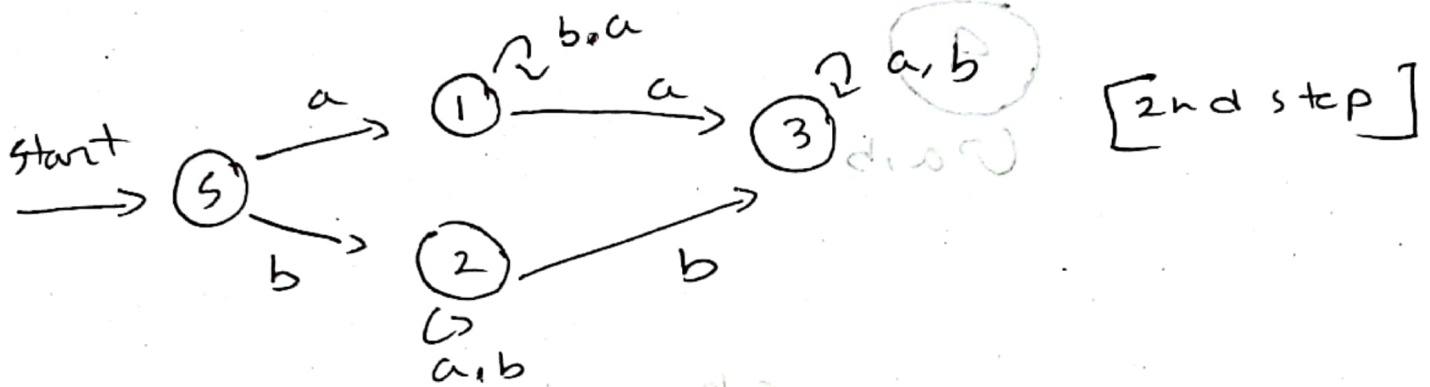
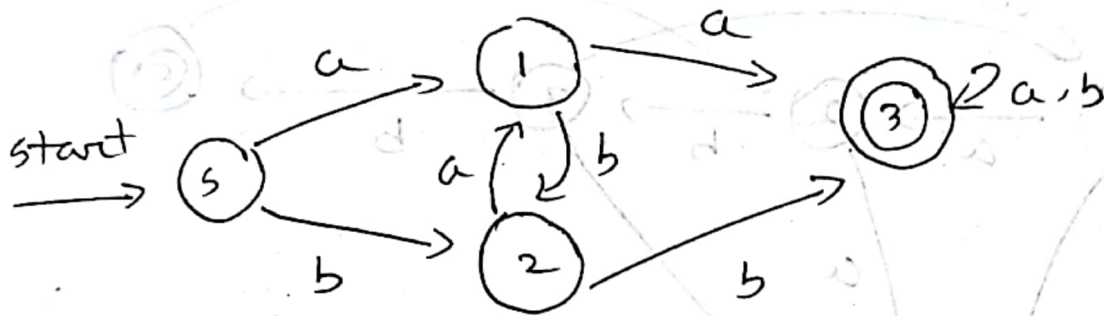
New Ques



[2nd step]

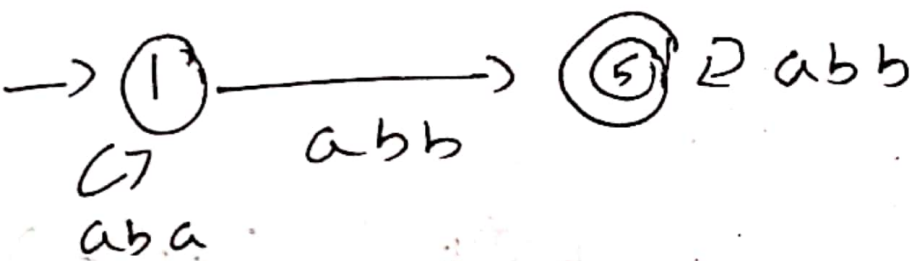
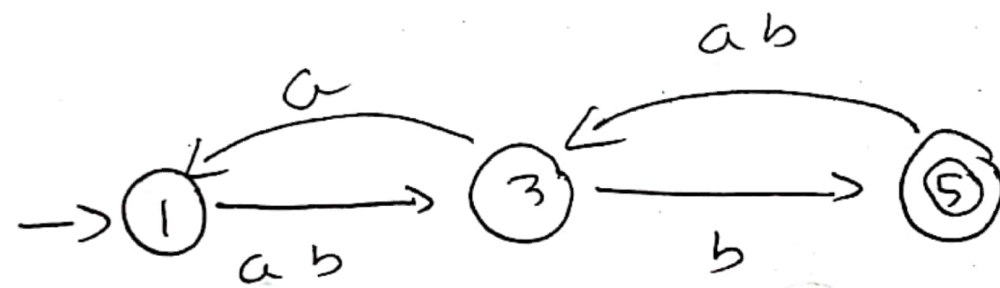
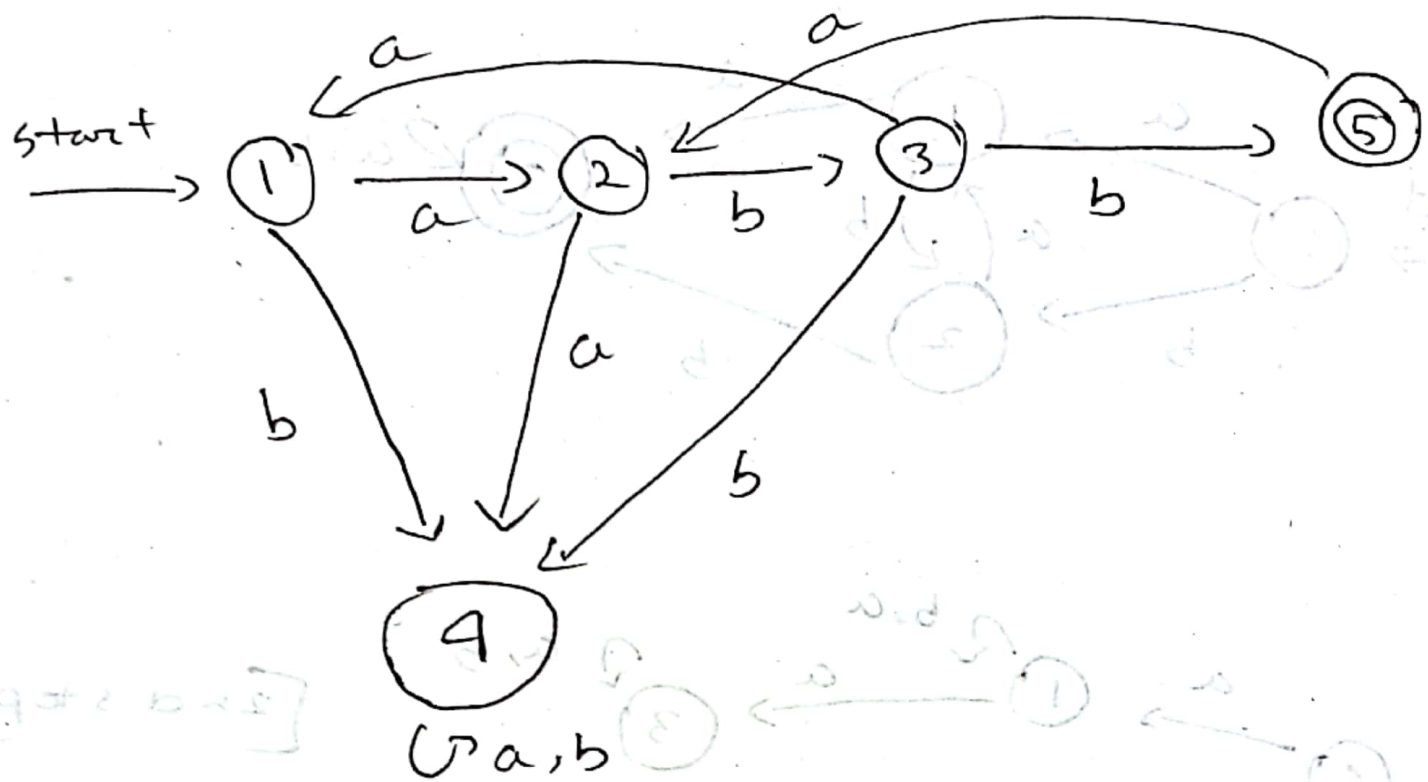


New Ques



$$(a(ba)^*a + b(ab)^*b)(a+b)^*$$

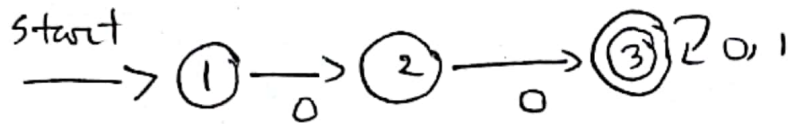
New Ques



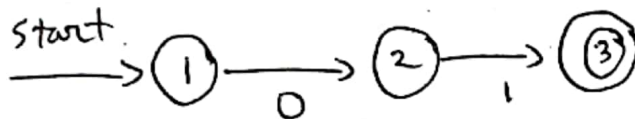
$$(aba)^* abb (abb)^*$$

RE to NFA

0.0



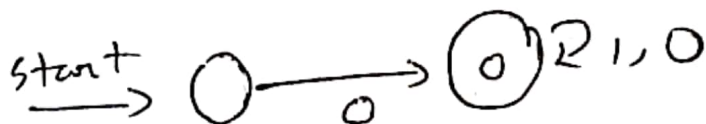
0.1



(0+1)



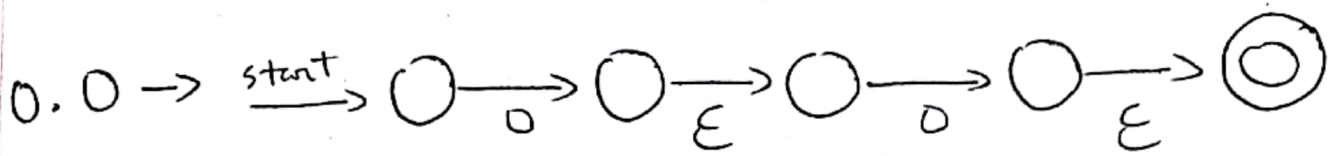
0.(1+0)*



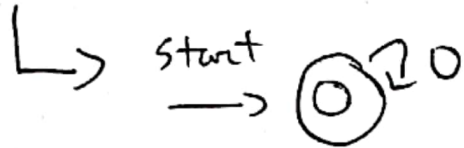
0.1*0



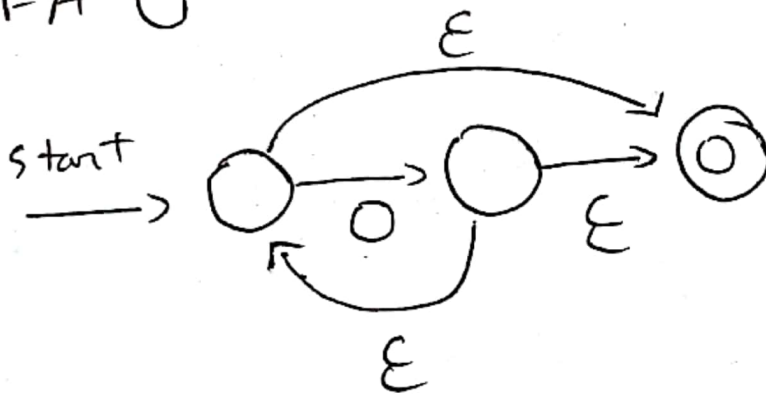
RE to ϵ NFA



Normal 0^*



ϵ NFA 0^*



$(10)^*$

